

Order Lepidoptera

Moths and butterflies

With over 165,000 species in 127 families, moths and butterflies are among the most diverse insects in the world. They have minute, overlapping scales on their bodies and wings, and mouthparts in the form of a proboscis. There is no clear-cut difference between butterflies and moths—and no scientific basis for separating them—but as a general rule, butterflies are brightly colored and day flying, with club-ended antennae, while moths are nocturnal and drab. In both, courtship often involves airborne scents and displays. The larvae, called caterpillars, have cylindrical bodies and chewing jaws. They develop by complete metamorphosis, often forming inside a silk cocoon.

Family Arctiidae

Tiger and ermine moths

Occurrence 11,000 spp. worldwide; in well-vegetated areas where host plants occur

Habitat Terrestrial

The hairy and heavy-bodied tiger moths are brightly colored. Ermine moths tend to be pale with black spots or patches and have a wingspan of $\frac{3}{4}$ –2 $\frac{3}{4}$ in (2–7 cm). Caterpillars are herbivorous and many are poisonous. Adults take nectar and liquids. Recent evidence groups actiid and noctuid moths together.



ARCTIA CAJA
Also called the garden tiger, this moth is highly unpalatable, and its bright, contrasting coloration advertises this to predators.

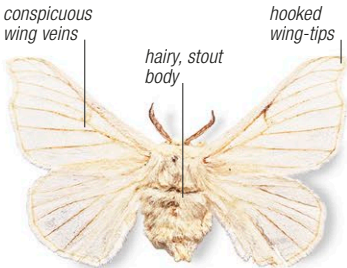
Family Bombycidae

Silk moths

Occurrence 100 spp. in tropical areas of Southeast Asia; on mulberry trees, other plants



Long valued as a source of silk, these stout, pale cream, gray, or brown moths have wings that span $\frac{3}{4}$ –2 $\frac{1}{4}$ in (2–6 cm). The smooth caterpillars have prolegs on all abdominal segments and usually a short, hornlike structure at the tail. Caterpillars of the commercial silk moth feed on mulberry trees; other species feed on fig and related plants. Pupation takes place inside a dense silk cocoon. Functional mouthparts are absent in adults, which do not feed.



BOMBYX MORI
The silk moth originated in China but is now cultivated in silk farms the world over. It is, however, extinct in the wild.

Family Geometridae

Geometer moths

Occurrence 30,000 spp. worldwide; on deciduous and coniferous trees, woody shrubs, herbaceous plants



The wings of these slender, generally nocturnal moths are rounded, with a span of $\frac{1}{2}$ –3 in (1.4–7.4 cm), and complex patterns of fine markings. The females of some species are wingless. These moths are colored brown or green for camouflage; coloration often varies between sexes. Females lay eggs singly or in groups, on bark, twigs, or stems of host plants. The characteristic looping motion of the caterpillars gives them the name “loopers” or “inchworms.” Many species are agricultural or forestry pests and can cause severe damage.



OOSPILA VENEZUELATATA
This green species is found mainly in the forests of Venezuela.

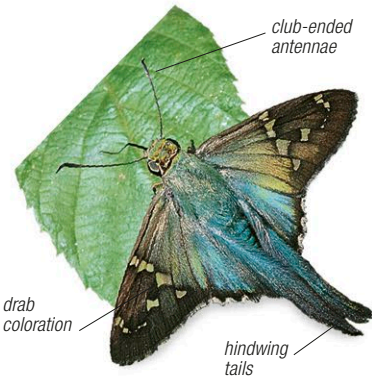
Family Hesperiidae

Skippers

Occurrence 3,500 spp. worldwide except New Zealand; in open habitats such as cultivated fields and grassland



Heavy-bodied and mothlike, these butterflies get their name from their rapid, darting flight. Most are drab brown with white or orange markings, but a few can be brightly colored. Their wings span $\frac{3}{4}$ –3 $\frac{1}{4}$ in (2–8 cm). The antennae end in a long, curved, and pointed club. Females lay single eggs on host plants. Caterpillars are herbivorous and feed at night.



URBANUS PROTEUS
The long-tailed skipper is common in parts of South and North America. It has characteristically long hindwing tails and an erratic flight pattern.

Family Lasiocampidae

Lasiocampids

Occurrence 2,000 spp. worldwide except New Zealand; on foliage of a wide range of deciduous trees



These insects, also known as lappet and eggar moths, are heavy-bodied and very hairy; most are yellowish brown, brown, or gray. The wingspan is mostly under 1 $\frac{1}{2}$ in (4 cm); females are bigger than males. Eggs are laid in a band of 100–200 around a twig of a host tree. The stout caterpillars may live

communally in silk tents or webs that catch the sun's rays. They pupate inside tough, egg-shaped, papery cocoons.



MALACOSOMA AMERICANUM
Also called the eastern tent caterpillar moth, this is an orchard pest. It inhabits forest and woodland areas in the northern USA and southern Canada.

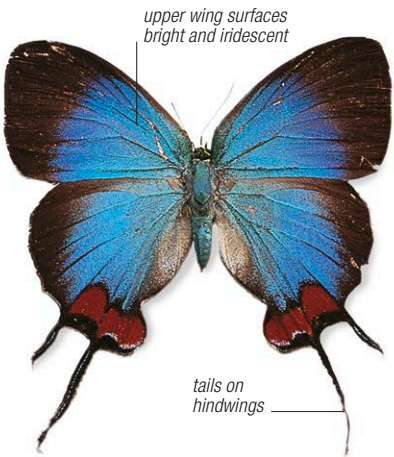
Family Lycaenidae

Blues, coppers, and hairstreaks

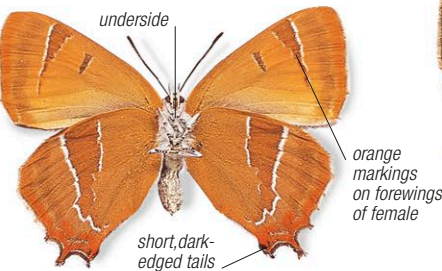
Occurrence 6,000 spp. worldwide, particularly in warmer regions; in association with ant nests or host plants



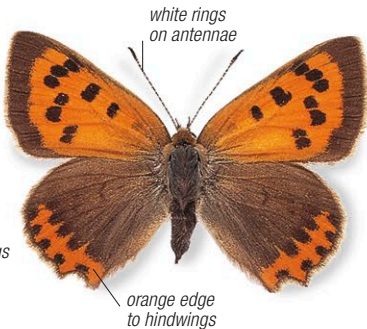
The butterflies in this family are generally small, with a wingspan of $\frac{1}{2}$ –2 in (1.5–5 cm), but they are notable for color differences between males and females. Males often have brilliantly colored, iridescent wing surfaces, while females are generally dull. Most species have squat caterpillars, which feed on plants or small insects. Some produce a special fluid that is eaten by ants—in return, the ants guard the caterpillar, and sometimes take it into their nest.



THECLA CORONATA
Also called the Hewitson's blue hairstreak, this species inhabits tropical forests of South America. It is one of the largest and most brilliantly colored members of the family.



THECLA BETULAE
The brown hairstreak butterfly inhabits woodland in Europe and temperate Asia. The caterpillars eat the leaves of sloe, plum, and birch trees.



LYCAENA PHLAEAS
The small copper butterfly is common in grassy areas up to 6,600 ft (2,000 m) in the Northern Hemisphere. The caterpillars feed on dock leaves.